THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

Explain software process models

① Waterfall Model

It is a linear and sequential software development approach in which each phase must be completed fully before the next begins. It is mainly used in engineering-type products where requirements are clearly defined

phases:

1. Requirement Analysis
All system requirements are gathered, documented, and approved.

2. System Design:
Architecture, data flow, interfaces, and system structure are designed.

3. Implementation (Coding)
Developers write code based on the design specifications

4. Integration & Testing
All modules are integrated & tested together. Bugs are fixed

5. Deployment
The finished product is released to the user.

6. Maintenance.
Updates, patches, and enhancement after deployment

Requirements → Design → Development → Testing → Deployment → Maintainence

THADOMAL SHA
TSEC
ENGINEERING COLLEGE

② Iterative Model (Spiral)

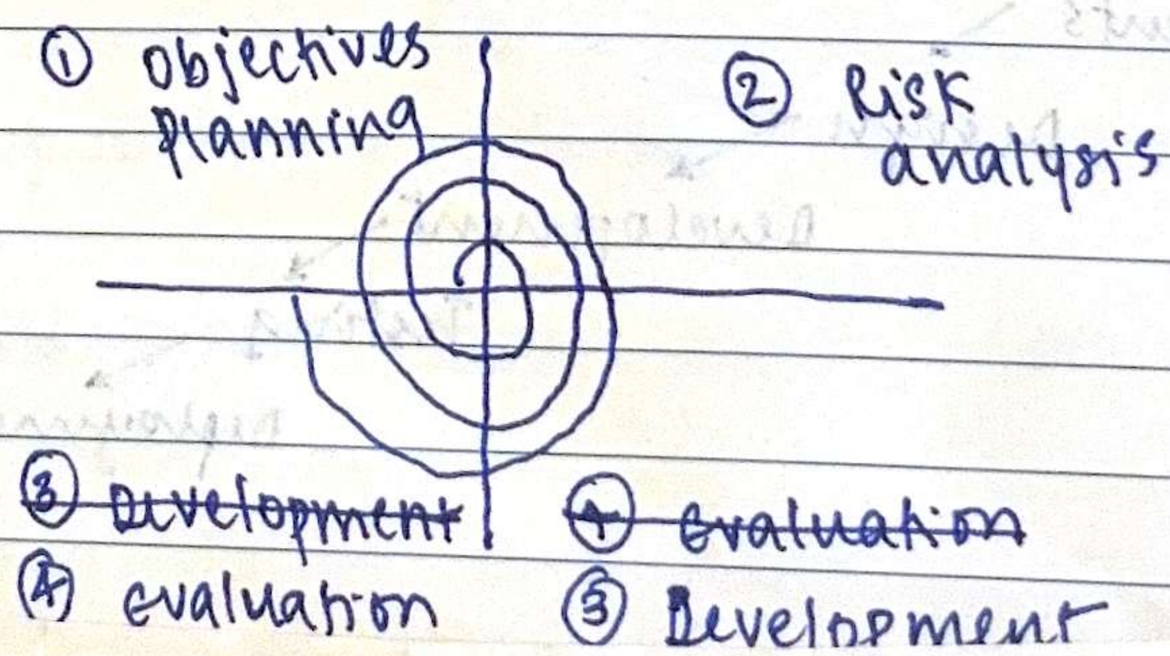
In the Iterative model, the project starts with a small sets of requirements and expands with each iteration. Instead of defining all requirements upfront, the project evolves based on feedback.

Spiral Model.

It is a risk driven development model. It combines Iterative ~~dev~~ development with systematic risk analysis. The project progresses in loops and each loop represents a development phase.

phases:

1. objectives planning
define goals, constraints, features and alternatives for that loop.
2. Risk analysis
Identify risks (technical, cost, schedule) → analyse → build prototypes if needed
3. engineering - design & development
develop selected features - design, coding, testing
4. Evaluation & Review
customer evaluates the output, planning for the next loop begins.



③ Incremental Model

The incremental model develops the system in small functional increments. Each increment adds new features. You design - implement - test a small part and repeat until the full product is complete.

phases:

① Analyse:

choose a small set of requirements to build this increment, only selected features are planned

② Design:

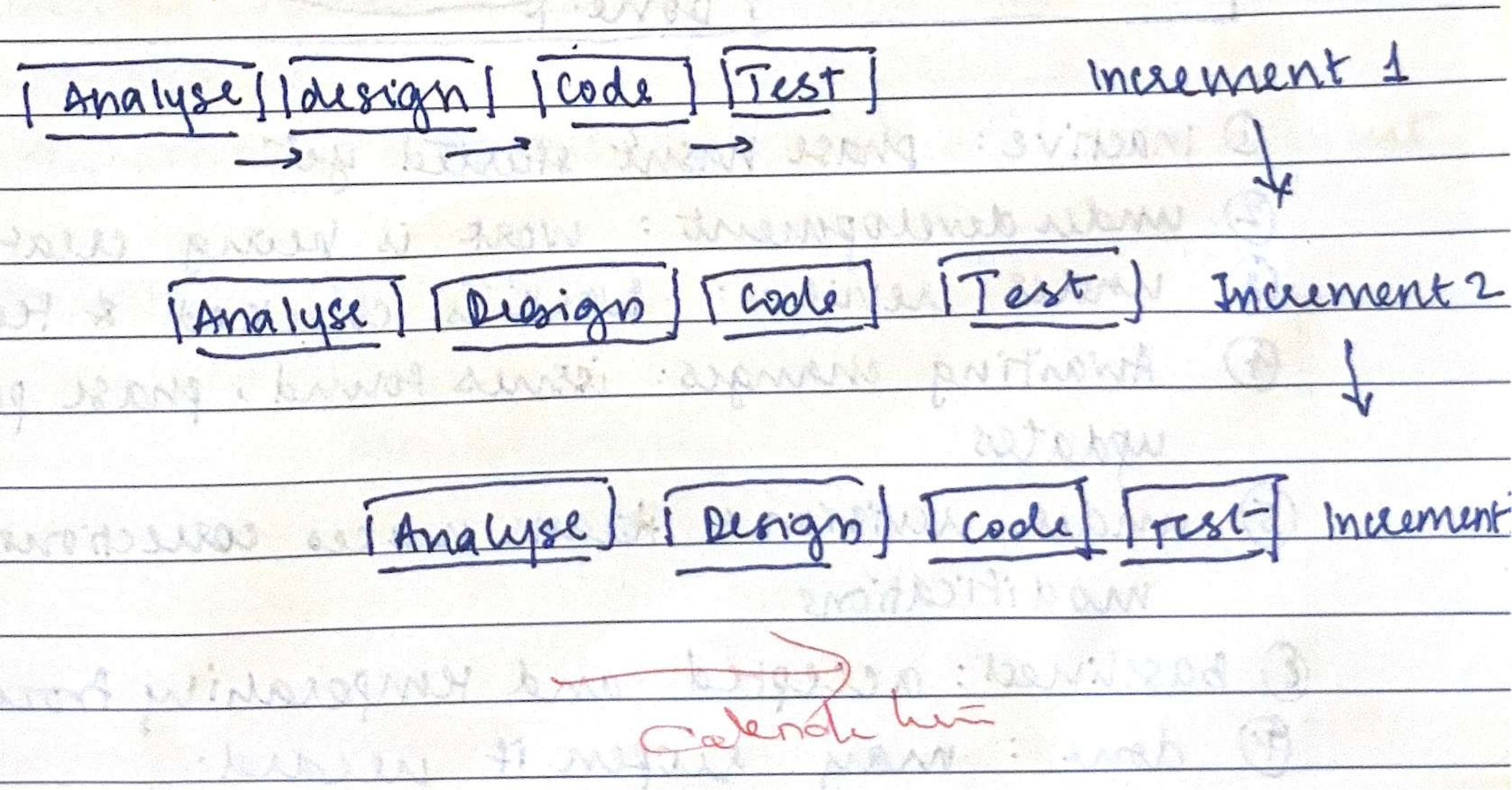
Design the architecture and components needed for just these features

③ Coding

Develop selected features, write & implement code.

④ Testing.

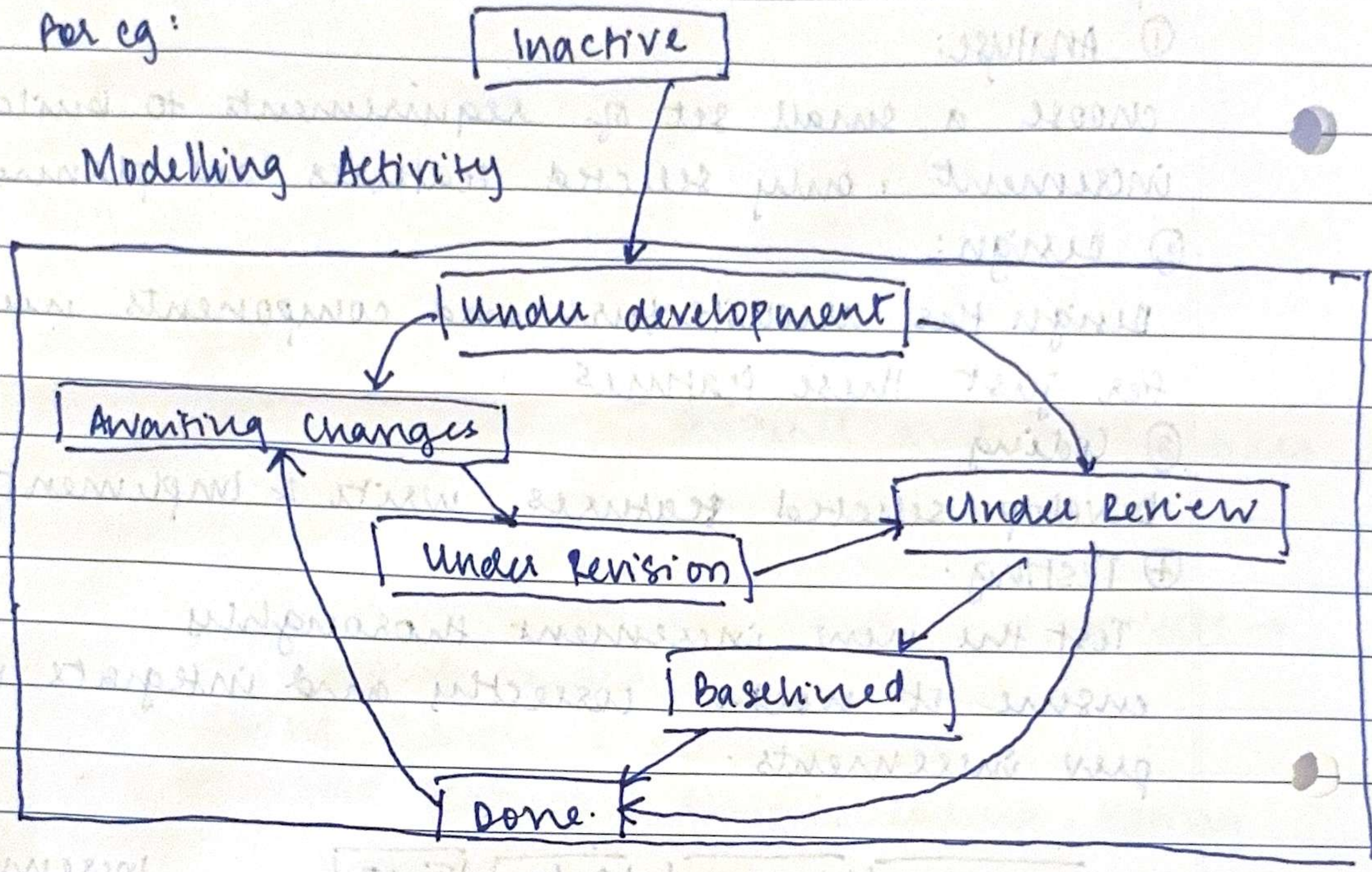
Test the new increment thoroughly ensure it works correctly and integrates with all prev increments.



④ concurrent model

This model allows all SDLC activities to run parallely. Each activity is one of several states. As soon as something changes in one activity the states of other activities also change. This makes the model flexible and ideal for project where requirements evolve continuously.

For eg:



① Inactive: phase hasn't started yet
② under development: work is being created
③ under review: work is checked & feedback is given
④ Awaiting changes: issues found, phase pauses for updates
⑤ under revision: team makes corrections or modifications
⑥ Baselined: accepted and temporarily frozen
⑦ done: may reopen if needed.